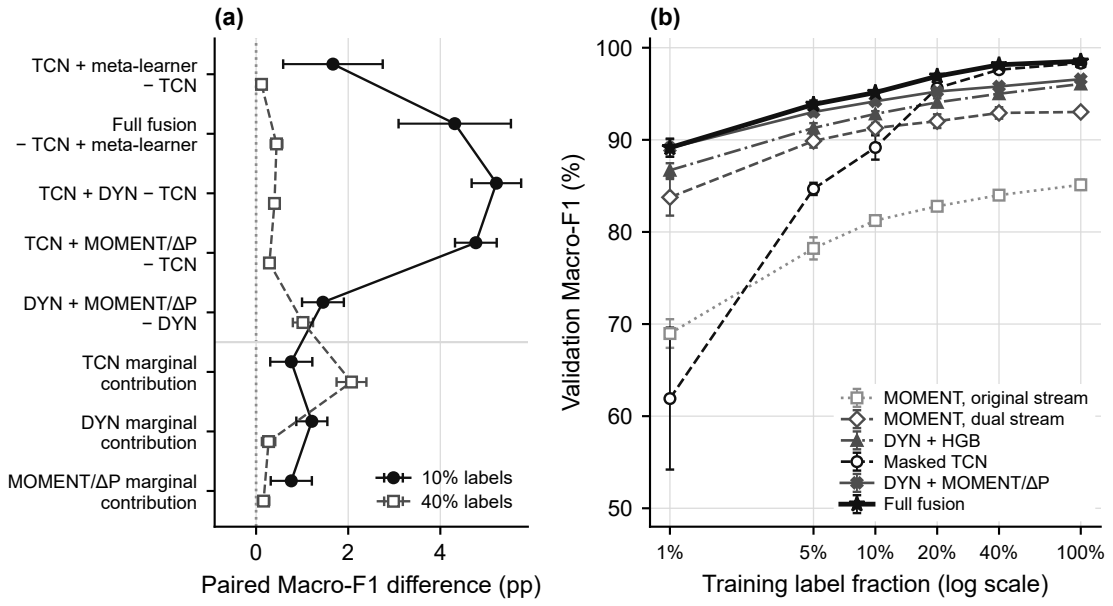

(a)
TCN + meta-learner − TCN
Full fusion − TCN + meta-learner
TCN + DYN − TCN
TCN + MOMENT/ΔP − TCN
DYN + MOMENT/ΔP − DYN
TCN marginal contribution
DYN marginal contribution
MOMENT/ΔP marginal contribution
10% labels
40% labels
0
2
4
Paired Macro-F1 difference (pp)
(b)
Validation Macro-F1 (%)
100
90
80
70
60
50
MOMENT, original stream
MOMENT, dual stream
DYN + HGB
Masked TCN
DYN + MOMENT/ΔP
Full fusion
1%
5%
10%
20%
40%
100%
Training label fraction (log scale)